Steps 1 and 2 compared with the Phase A document

What was changed, what was added, and what still has to be changed. Project 26-2-R-2 - Niran Czuckermann and Shahar Bar. Only step 1 (cleaning) and step 2 (breaking the problem into parts) are covered here.

1. In short

- **Step 1 (cleaning) is unchanged** and follows Phase A. It is repeated in section 2 only for completeness.
- **Step 2 (breaking the problem into parts):** after checking every difference from Phase A, only **two** are kept: the closeness formula and the second forest. Everything else is Phase A, a clarification, or an optional extra (section 3).
- **Closeness is the big one.** Phase A's formula returns 0 for every paper on a citation network. Section 4 gives the formula before and after, an example, the measurements and a decision record.
- Section 5 lists what has to be done.

2. Step 1 - Cleaning the database

Phase A, chapter "Anticipated Challenges - Dataset Noise": "the data cleaning pipeline will include removal of duplicate edges, elimination of self-loops, validation of temporal consistency, and filtering of isolated nodes." The unit-test plan repeats the same list.

Topic	Phase A says	What we did	Change?
Dataset	Cora, "under preliminary review and subject to refinement"	PubMed-Diabetes: 19,717 papers, 44,338 citations (the dataset given for the project). Used as is, no sampling	**Yes.** Phase A text is outdated
Duplicate edges	Remove	Removed. **0 found**	No
Self-loops	Remove	Removed. **3 found** (papers 10751750, 9221759, 1586783)	No
Temporal consistency	Validate (implies checking publication years)	The dataset has **no dates**. Checked structurally: citations must not form loops. **11 pairs of papers cite each other**, which is impossible in time. For each pair we kept the edge from the higher PubMed ID to the lower one (higher ID = newer paper, an assumption). Result: 0 cycles, the graph is a DAG	**Yes.** Method differs, plus one assumption
Isolated nodes	Filter	Filtered. **0 found**	No
Edges to unknown papers	Not mentioned	Added as a safety check. 0 found	Addition
Word-feature columns	Not mentioned	The 500 TF-IDF columns are dropped; the class label is kept but not used; the model uses graph structure only	Addition
Edge direction	"Directed edges represent citations" (direction not spelled out)	A cites B is stored as edge A -> B	Clarification

Final graph: 19,717 papers, 44,324 citations (14 fewer than the raw file), one connected piece, no loops.

What Phase A did not expect

- **80 % of papers cite nothing** (15,840 of 19,717) and 2,047 are never cited. The dataset contains only citations that point to papers inside it. This matters later for closeness and for PageRank (see step 2).
- **The ID rule is only a proxy.** We checked it afterwards: 89.6 % of all citations go from a higher ID to a lower one, so the rule is right most of the time, not always. It affects 11 edges out of 44,338.
- Phase A also mentions "false positive" and "false negative" citations. We cannot detect either in this dataset; they are only simulated later by adding and removing edges.

3. Step 2 - Every difference from Phase A, and what is kept

Rule used: we stay with Phase A unless Phase A cannot work on this data or cannot answer its own question. Each difference was checked against that rule. **Only two differences are kept as changes.**

#	Point in step 2	Phase A	Ours	Verdict
1	Closeness formula	CL(v) = (N-1) / sum of d(v,y), distances going out from v	Wasserman-Faust, distances coming in to v (section 4)	**KEEP - critical.** Phase A's formula gives 0 to every paper
2	Number of forests	One forest; PageRank is an input and also the target	Two forests: with PageRank (= Phase A as written) and without	**KEEP - important.** With PageRank as input the forest copies it (99.8 % of its importance)
3	Betweenness	NetworkX or igraph, exact formula	Exact, with igraph	Not a change. We first planned sampling, measured it (Spearman 0.55 with the exact one), and returned to Phase A
4	k-core	Standard definition (undirected)	Computed on the undirected graph	Not a change
5	Degree, Min-Max scaler, zero-mean rule in CV, seeds, training on the original graph	As defined	Same	Identical
6	Edge direction (A cites B = A -> B)	Not spelled out	Written down	Clarification, not a change
7	Second way of adding edges (newer -> older only)	Random edges	Phase A version kept and run; the second version is extra	Not a step 2 change (belongs to step 5). Optional extra; can be dropped without touching anything else
8	80/20 hold-out check of the forest	Not mentioned	Extra sanity check, reported only	Not a change to the method. Dropped from the change list
9	Adjacency matrix (requirement 2)	Required	No code builds one	A gap against Phase A. Closing it moves us closer to Phase A (section 5)

4. Closeness centrality - the big change in detail

4.1 Phase A: formula and what happens on this data

$$\mathrm{CL}(v) = (N-1) \,/\, \Sigma\, d(v,y) \qquad \text{sum over all papers } y \neq v$$

N is the number of papers, and d(v, y) is the length of the shortest path **from v to y**, following citations forward (v cites something that cites something ... that cites y). The text of Phase A says the same in words: "the average shortest path distance from a node to all other nodes".

If some paper y cannot be reached from v, then d(v, y) = infinity, the sum is infinite and CL(v) = 0. A citation network is a directed graph with no loops: a paper cites older papers, and the oldest papers cite nothing. So **no paper can reach all the others**, and the formula returns 0 for every paper. We measured it: 0 for 19,717 of 19,717 papers, one single value.

A feature that is the same for every paper tells the forest nothing, and ranking by it is meaningless. The formula has to be extended to pairs with no path. (Phase A's test plan already expects this case: "returning 0 when no finite path exists". It does not say how to do it for the sum.)

4.2 What we use: formula

Let R(v) be the set of papers from which v can be reached (the papers that cite v, directly or through a chain of citations), and r(v) = |R(v)|. Then

$$\mathrm{CL}(v) = [\, r(v) \,/\, \Sigma\, d(u,v) \,] \cdot [\, r(v) \,/\, (N-1) \,] \qquad \text{sum over } u \text{ in } R(v);\ \ \mathrm{CL}(v) = 0 \text{ if } r(v) = 0$$

- **First bracket:** the inverse of the average distance, but only over papers that have a path. This is Phase A's idea (short average distance = high closeness), applied where distances exist.

- **Second bracket:** the share of the network that has a path to v. It is the Wasserman-Faust correction. Without it, a paper cited by a single paper would look perfect (section 4.4).
- **Distances go in, not out.** d(u, v) is the distance **to** v, from the papers that cite it.

Link to Phase A. If every other paper reaches v, then r(v) = N - 1, the second bracket is 1, and the first is exactly (N-1) / sum of distances. So our formula equals Phase A's formula whenever Phase A's formula is defined (apart from the direction of the distances). It only adds a rule for the cases where Phase A's formula breaks.

4.3 Small example

Four papers: A cites B and C; B cites C; C cites D. Edges: A->B, A->C, B->C, C->D. N = 4.

Paper	Phase A (distances out)	Ours (distances in)	What it is
A	d to B, C, D = 1, 1, 2, sum 4 -> CL = 3/4 = **0.75**	nobody reaches A: r = 0 -> **0**	newest paper, cited by nobody
B	A is not reachable -> infinite -> **0**	reached by A (distance 1): (1/1)(1/3) = **0.33**	cited once
C	D reachable but not A, B -> infinite -> **0**	reached by A, B (distances 1, 1): (2/2)(2/3) = **0.67**	cited twice
D	reaches nobody -> **0**	reached by C (1), A (2), B (2): (3/5)(3/3) = **0.60**	cited by C, which all others reach

Even when we add a rule for unreachable pairs, the outgoing direction gives the highest closeness to the paper that nobody cites (A) and zero to the papers that others build on. Ours ranks C and D at the top, and A at zero, which is what an importance measure for citations should do. This also matches Phase A's own sentence: closeness reflects "how rapidly a node's influence reaches across the network". A paper's influence travels from the cited paper to the papers that cite it, that is, along incoming distances.

4.4 Measured on the real graph

Version	Papers with CL = 0	Different values	Spearman with PageRank
Phase A as written (distances out)	19,717 (100 %)	1	undefined
Out, only reachable papers, no correction	15,840 (80.3 %)	1,908	-0.27
Out + Wasserman-Faust	15,840 (80.3 %)	2,141	-0.27
In, only reachable papers, no correction	2,047 (10.4 %)	2,481	+0.02
In + Wasserman-Faust (used)	**2,047 (10.4 %)**	**2,814**	**+0.73**

- **Why not outgoing:** 15,840 papers (80 %) cite nothing inside the dataset and reach nobody, so they all get 0. The correlation with PageRank is negative: papers with long reference lists score high, heavily cited papers score low.
- **Why the correction:** without it the correlation with PageRank drops from +0.73 to +0.02, and a paper cited once from one step away gets the maximum score 1.0.
- **Cost of the choice:** in step 4 the forest predicted PageRank equally well with either direction. The direction changes what the feature means, not the accuracy.

4.5 Decision record (also written as D17 in decisions.md)

Item	Content
Decision	Closeness = Wasserman-Faust with incoming distances
Replaces	Appendix B closeness formula (N-1) / sum d(v,y), outgoing distances
Reason	The Phase A formula gives 0 to all 19,717 papers on a citation DAG; outgoing distances with a correction still give 0 to 80 %; incoming distances follow the direction in which a paper's influence travels
Evidence	Table 4.4, reproduced by a script on the cleaned graph; the igraph version is tested against networkx
Risk	Phase A's text says "from a node to all others". An examiner may read it as outgoing. Answer: the formula is undefined there, and the incoming version keeps the stated meaning
Alternative	Switch to outgoing: change one argument (mode in -> out) in features.py and rerun everything after step 2. About 15 minutes of computing; the feature would then be 0 for 80 % of the papers
Status	**Needs the advisor's approval and the team's agreement**

5. What has to be done

#	Action	Type
1	Advisor: approve closeness (section 4) and the two forests	decision
2	Report: replace the closeness formula in Appendix B with the one in 4.2	text
3	Report: feature vector and flow chart show both forests (with and without PageRank)	text
4	Code: add a function that builds the adjacency matrix (requirement 2) with a test. About 15 minutes, no effect on results	code, optional
5	Decide whether to keep the second way of adding edges (row 7 in section 3). Keeping it costs nothing	decision, optional

This list replaces the longer one in the earlier version of this file. Step 1 (cleaning) is unchanged.